

Mastering the Model Context Protocol (MCP)

Building the "USB-C" for AI Agents

The Problem - The “m x n” nightmare

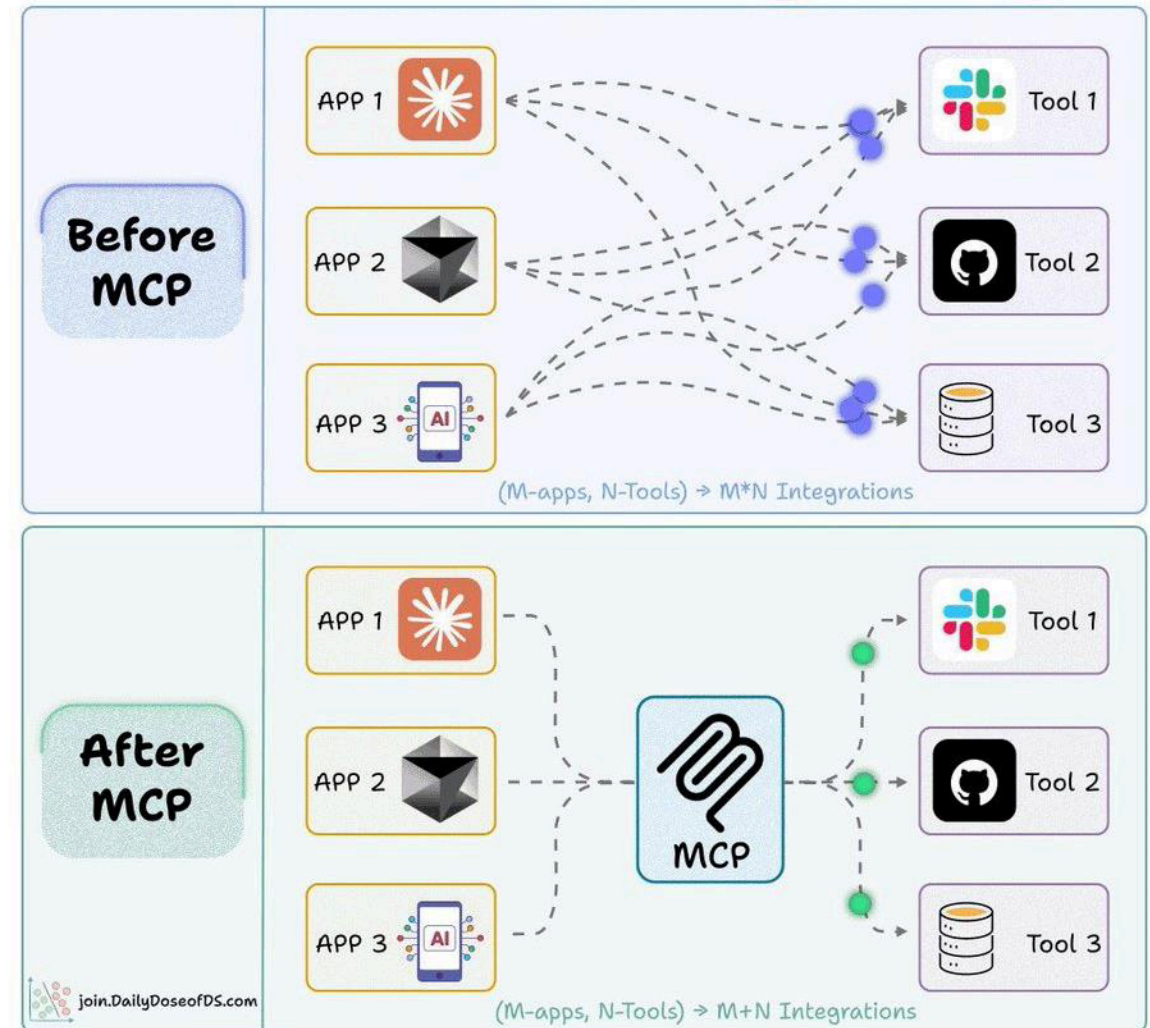
Every time a new AI comes out, developers have to rewrite custom "connectors" for every data source.

The Solution - MCP?

An open-source standard introduced by Anthropic to unify how AI connects to data.

Just as one cable connects your phone, laptop, and camera to any charger, MCP allows **any** AI to talk to **any** data source.

Works with Claude Desktop, IDEs (like Cursor/Zed), and custom Python scripts.



Project Overview: "The SQL-Talker"

Create an AI-powered adapter that lets any LLM safely "chat" with a MySQL database using natural language.

Project Goal & Objectives: Bridge the gap between "Raw Data" and "Smart AI" (Natural Language) using a universal connector.

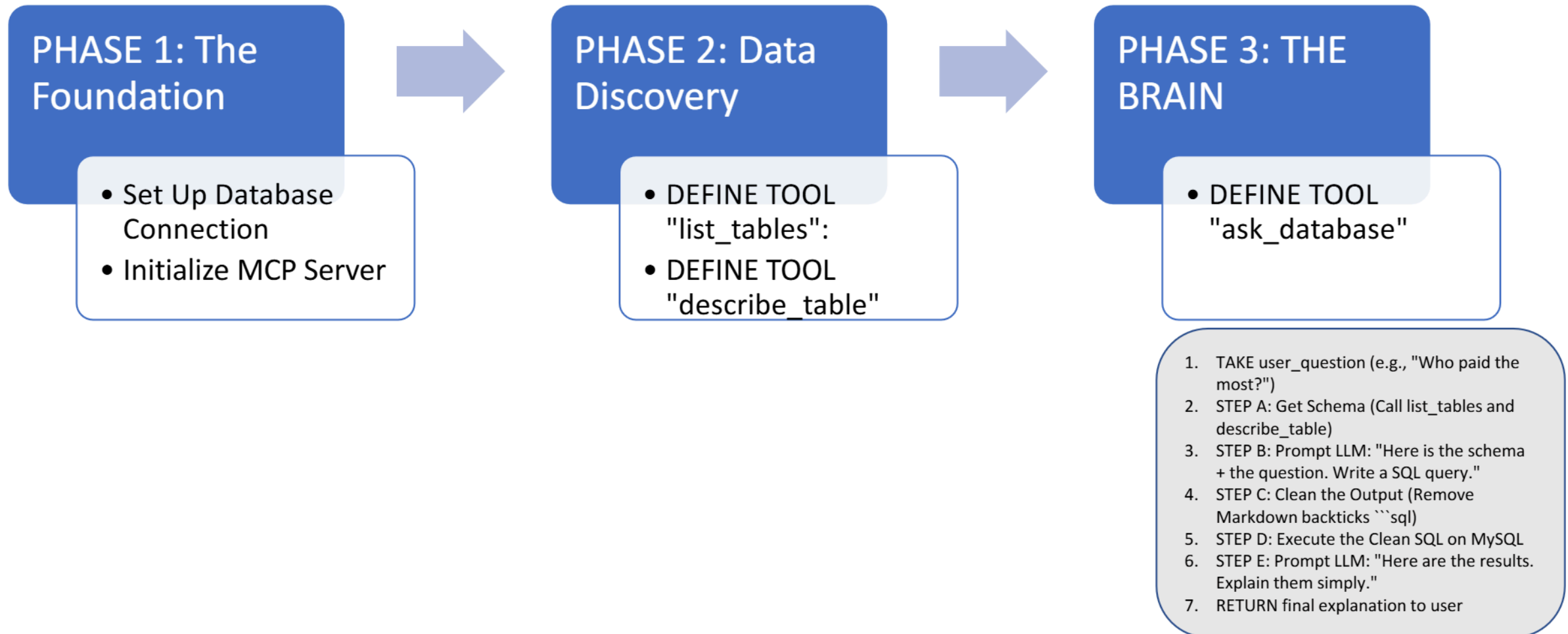
Learning Objectives:

- Understand the **Client-Server-Host** architecture of MCP.
- Master **NL2SQL** (Natural Language to SQL) translation.
- Learn to build **Agentic Tools** that an AI can discover and execute automatically.
- Practice **Data Cleaning** to ensure AI-generated code doesn't crash production systems.

Core Features (The "What it Does")

- Automatic Discovery: The AI can "see" what tables exist without being told.
- Contextual Memory: The AI remembers your table structure (schema) to write accurate queries.
- Safe Execution: Only SELECT queries are allowed, protecting the database from accidental deletion.
- Human-Readable Answers: The raw data rows are "translated" back into a friendly summary for the user.

Pseudocode



THANK YOU!